Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. | *Subject: Electrical Circuit Analysis* | *Time: 03 Hours*
Code: 100306 | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Kirchhoff's Current law is based on law of conservation of
(i) energy (ii) momentum (iii) mass (iv) charge

(b) Determine the current if a 20 coulomb charge passes a point in 0.25 seconds
(i) 10 A (ii) 20 A (iii) 2 A (iv) 80 A

(c) Comment on the linearity of y[n] = n*x[n].
(i) Linear (ii) Only additive (iii) Not scalable (iv) Non linear

(d) Maximum power transfer occurs at a
(i) 100% efficiency (ii) 50% efficiency
(iii) 25% efficiency (iv) 75% efficiency

(e) The inverse laplace transform of is $\frac{1}{s} - \frac{e^{-as}}{s}$
(i) U(t)-U(t-a) (ii) U(t) (iii) U(t-a) (iv) Zero

(f) The parameter A of a two port network is equal to
(i) V_2 / V_1 with $I_1 = 0$ (ii) V_1 / I_2 with $V_2 = 0$
(iii) I_1 / I_2 with $V_2 = 0$ (iv) V_1 / V_2 with $I_2 = 0$

(g) A system is linear if it satisfies
(i) Principle of superposition (ii) Principle of homogeneity
(iii) Both (i) & (ii) (iv) Only (i)

(h) Given F(S) = (S+2)/S(S+1), the initial and final values of f(t) will be respectively
(i) 1, 2 (ii) 2, 1 (iii) 1, 1 (iv) 2, 2

(i) Time constant of series RL circuit is
(i) L/R (ii) RL (iii) R/L (iv) None of the above

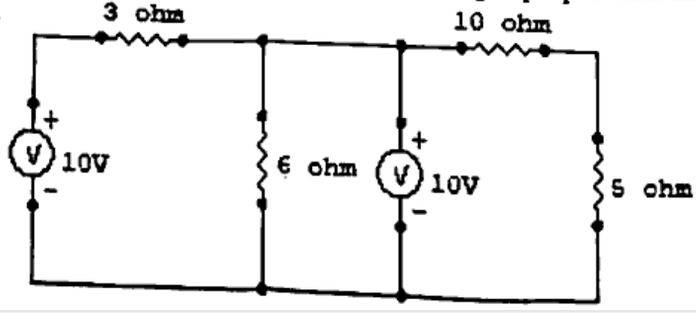
(j) The transmission parameters are also known as:
(i) hybrid parameters (ii) A B C D parameters
(iii) impedance parameters (iv) admittance parameters

Q.2 (a) A series R-L circuit has R= 2 Ω and L=2H. A ramp input r(t) is applied at t=1sec. Obtain an expression for current through the circuit. **[10]**

(b) Define time constant. What the significance of time constant? **[4]**

Q.3 (a) State Superposition theorem explaining the conditions in which it is valid. **[6]**

(b) In the network shown in figure below, find the value of current flowing through the 6 ohm resistance in the circuit given below using superposition theorem. **[8]**

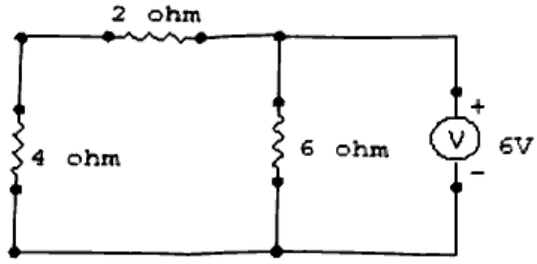


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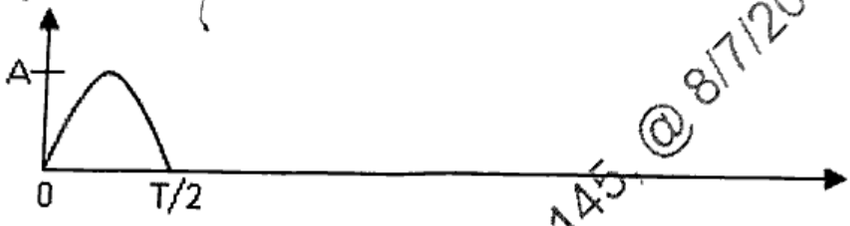
Q.4 (a) State and explain Thevenin's Theorem. Show the Thevenin's equivalent representation of any circuit. [7]

(b) For the circuit given below, determine the current flowing through the 4 ohm resistance by Thevenin's theorem. What are the values of Thevenin's resistance and Thevenin's voltage for this circuit. [7]



Q.5 (a) Calculate the Laplace Transform of the function $f(t) = e^{-t} \sin wt$ [6]

(b) Define Gate function? Using Gate function find out the Laplace Transform of a single Sine wave shown below. [8]



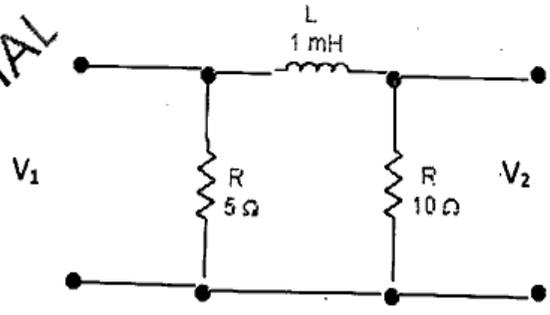
Q.6 (a) Calculate the impedance, resistance, power and power factor of a circuit whose expression for voltage and current are given by- [10]

$V = 100 \sin (wt + 600) - 50 \sin (3wt - 300)$ volt.

$I = 10 \sin (wt + 600) - 5 \sin (3wt + 300)$ amp.

(b) A 400 mH coil of zero resistance is connected to an AC circuit in which 6 mA current is flowing. Find out the voltage across the coil if the frequency is 100 Hz. [4]

Q.7 Obtain the Y & Z parameter of the network shown below. Write the unit of all individual parameters. [14]



Q.8 (a) Explain frequency response. Why magnitude and phase plots are necessary for frequency response analysis. [7]

(b) What is Q-factor? Calculate the quality factor of a series RLC circuit with L = 2 H, C = 2 μF and R=20 ohm. [7]

Q.9 Write short notes on *any two* of the following: [7 x 2 =14]

(a) Norton's Theorem
(b) Reciprocity Theorem
(c) Hybrid Parameters of a 2-port network
(d) Duality and Dual Network.

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